



Modelling Financial Markets

Seminar 16: Digital Assets and DeFi — Hands-on



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ ETF: Exchange-Traded Fund
- ▣ ETHA: iShares Ethereum Trust ETF (ticker)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ KPSS: Kwiatkowski–Phillips–Schmidt–Shin test
- ▣ LVR: Loss-Versus-Rebalancing
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ USDC: USD Coin (stablecoin issued by Circle)
- ▣ XRP: XRP (the native token of the XRP Ledger)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Questions, AI Critique, Project Seed

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the no-arbitrage band, the optimal arbitrage and the LVR of constant-product and weighted pools
- ▣ Model a stablecoin peg as a threshold AR and test linearity with a bootstrap sup-test
- ▣ Test for a break at an unknown date and choose between level and difference estimators
- ▣ Report every estimate with a valid interval: block bootstrap, HAC, Dimson, Forbes–Rigobon

- ▣ Background: Lecture 16; textbooks [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** eight computational exercises with inference (B1–B8)
- ▣ **Part C:** an open question, a seed for the team project (C1), and an AI answer to critique (C2) and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Crypto trades 7 days a week: annualise with 365 days; joint analyses with shares use common days (prices joined first)
- Report every estimate with its uncertainty: bootstrap interval or robust standard error

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (index and divisor), A3 (fee band and optimal arbitrage), A5 (weighted-pool impermanent loss), A7 (threshold AR algebra)
- Part B: B1 (Bitcoin volatility and the weekend effect), B3 (USD Coin threshold AR), B5 (IBIT: Dimson beta and fee drift), B7 (Strategy: break at an unknown date)

Proposed: you solve them

- Part A: A2 (replacing a constituent; model: A1), A4 (arbitrage in a weighted pool; model: A3), A6 (LVR by $\text{It}\hat{o}$; model: A5), A8 (what a linear AR(1) estimates; model: A7)
- Part B: B2 (Ether and Solana; model: B1), B4 (Tether and Dai; model: B3), B6 (ETHA; model: B5), B8 (Coinbase; model: B7)
- Part C: C1, open question and project seed; C2, critique of an AI answer (Sharpe ratio and stablecoins); C3, crypto momentum after 2018 (model: B1, B7)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter16_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`
- Series: Bitcoin, Ether, Solana; Tether, USD Coin, Dai; IBIT, ETHA, Coinbase, Strategy; S&P 500, Nasdaq 100 (QQQ), gold (GLD), long Treasuries (TLT); daily closes up to 18 September 2026
- Crypto: 7 days a week; shares and ETFs: weekdays, prices adjusted for dividends and splits
- Weekly returns: Friday closes on common days, then log returns

Formulas Used Today

- ▣ Laspeyres index: $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$; at a reallocation $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
- ▣ Constant product $xy = k$, fee f , $\gamma = 1 - f$: $\Delta y = y\gamma\Delta x / (x + \gamma\Delta x)$ Adams, Zinsmeister & Robinson (2020)
- ▣ Impermanent loss $IL(r) = 2\sqrt{r}/(1+r) - 1$; loss versus rebalancing $\sigma^2/8$ a year Milionis et al. (2022)
- ▣ Arbitrage band of a redeemable token: $1 - c_r \leq P \leq 1 + c_m$ Lyons & Viswanath-Natraj (2023); threshold AR with band c Balke & Fomby (1997), linearity test Hansen (1996)
- ▣ Break at an unknown date: sup-Wald Andrews (1993), CI for the date Bai (1997); moving-block bootstrap with blocks of 20–21 days Künsch (1989)
- ▣ Hedge and safe haven Baur & Lucey (2010): $r_t^{\text{BTC}} = a + b r_t^{\text{S\&P}} + \sum_q c_q r_t^{\text{S\&P}} 1\{r_t^{\text{S\&P}} \leq q\} + u_t$

A1: A Three-Coin Index and Its Divisor [Solved]

Task

▣ Data and assumptions

- ▶ Three coins A, B, C on the base day: prices $P_0 = (60\,000; 3\,000; 150)$ USD and circulating supplies $Q = (19.5; 120; 500)$ million
- ▶ Laspeyres index $I_t = \sum_i P_{i,t} Q_{i,\tau} / D_\tau$ with quantities fixed within the month (lecture, Section 3)

▣ Tasks

- ▶ (a) Compute the market values, the weights and the divisor D_0 that sets $I_0 = 1000$.
- ▶ (b) At month end $P_1 = (66\,000; 2\,700; 180)$: compute I_1 and check it as a weighted average of the three returns.
- ▶ (c) At the reallocation the supplies become $(19.52; 120.1; 560)$ million: compute the new divisor D_1 and the new weights.

▣ Report

- ▶ D_0 , I_1 , D_1 and both sets of weights, with every intermediate product

▣ Interpretation question

- ▶ Why must the divisor change at a reallocation, and what would an investor see without the change?

A1: A Three-Coin Index and Its Divisor (Solution 1/2) [Solved]

- (a) Market values $M_i = P_{i,0} Q_i$: 1 170, 360, 75 billion USD, total 1 605 billion.
- Weights $w_i = M_i / \sum_j M_j$: 72.9%, 22.4%, 4.7%; divisor $D_0 = \sum_j M_j / 1000 = 1.605 \cdot 10^9$.
- (b) $I_1 = \sum_i P_{i,1} Q_i / D_0 = 1059.81$; the returns are +10%, -10%, +20%.
- Check: $72.9\% \cdot 10\% - 22.4\% \cdot 10\% + 4.7\% \cdot 20\% = 5.98\% = I_1 / 1000 - 1$.

A1: A Three-Coin Index and Its Divisor (Solution 2/2) [Solved]

- (c) New basket value at month-end prices: $\sum_i P_{i,1} Q_i^{\text{new}}$; the new divisor keeps the level:
 $D_1 = \sum_i P_{i,1} Q_i^{\text{new}} / I_1 = 1.6167 \cdot 10^9$.
- New weights: 75.2%, 18.9%, 5.9%; the level stays 1059.81 on the switch day.
- Interpretation: the divisor absorbs changes in the basket that are not price moves; without it the index would jump by the value of the added coins, a fake return.

A2: Replacing a Constituent [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ The index of A1 after the reallocation in (c): level $I = 1059.81$, divisor $1.6167 \cdot 10^9$
- ▶ On the same day coin C is delisted and coin D enters with $P_D = 0.5$ USD and $Q_D = 40$ billion coins

□ Tasks

- ▶ (a) Compute the market value of the old basket and of the new basket on the switch day.
- ▶ (b) Derive the new divisor that keeps I continuous and compute it.
- ▶ (c) Compute the index level the day after if the divisor were not changed and prices did not move.

□ Report

- ▶ Both basket values, the new divisor and the level of (c)
- ▶ Model: A1

□ Interpretation question

- ▶ Why does the divisor fall here, and who would be misled by a fake loss?

A3: The No-Arbitrage Band of a Pool with Fees [Solved]

Task

□ Data and assumptions

- ▶ Constant-product pool $xy = k$ with $x = 1\,000$ Ether and $y = 3\,000\,000$ USD Coin, so the pool price is $p = y/x = 3\,000$ Adams, Zinsmeister & Robinson (2020)
- ▶ Fee $f = 0.3\%$ charged on the input, $\gamma = 1 - f$; the outside market price is P ; no gas cost, one trade

□ Tasks

- ▶ (a) Derive the marginal cost of buying and the marginal revenue of selling Ether in the pool, and show that no trade is profitable while $\gamma P \leq p \leq P/\gamma$.
- ▶ (b) For $P = 3\,300$ write the profit $\pi(\Delta y)$ of buying Ether with Δy USD Coin, solve the first-order condition and compute Δy^* ; the fee stays in the pool (Uniswap v2): compute the actual reserves and the new pool price y/x .
- ▶ (c) Compute the arbitrage profit with and without the fee.

□ Report

- ▶ The band, Δy^* , Δx , the new price and both profits

□ Interpretation question

- ▶ Where does arbitrage leave the pool price, and who pays the arbitrage profit?

A3: The No-Arbitrage Band of a Pool with Fees (Solution 1/2) [Solved]

- (a) Buying Δx Ether: $(x - \Delta x)(y + \gamma \Delta y) = k$ gives $\Delta y = \frac{1}{\gamma} \left(\frac{k}{x - \Delta x} - y \right)$, so the marginal cost at $\Delta x = 0$ is $\frac{k}{\gamma x^2} = p/\gamma$.
- Selling Ether: $(x + \gamma \Delta x)(y - \Delta y) = k$ gives a marginal revenue γp .
- Buying pays iff $p/\gamma < P$, selling pays iff $\gamma p > P$; neither holds when $\gamma P \leq p \leq P/\gamma$, here $[3290.10, 3309.93]$ for $P = 3\,300$.

A3: The No-Arbitrage Band of a Pool with Fees (Solution 2/2) [Solved]

- (b) $\pi(\Delta y) = P[x - k/(y + \gamma\Delta y)] - \Delta y$; $\pi'(\Delta y) = Pk\gamma/(y + \gamma\Delta y)^2 - 1 = 0$ gives the effective reserve $y^e = y + \gamma\Delta y^* = \sqrt{\gamma Pk}$.
- Substituting $k = 3 \cdot 10^9$, $\gamma = 0.997$, $P = 3300$: $y^e = 3141703$, $x' = k/y^e = 954.90$; $\Delta y^* = (y^e - y)/\gamma = 142130$ USD
Coin buys $\Delta x = 45.10$ Ether.
- The fee stays in the pool: actual reserves x' and $y' = y + \Delta y^* = 3142129.75$, so the new pool price is $p' = y'/x' = 3290.55$, above $y^e/x' = \gamma P = 3290.10$; k grows with the retained fee.
- (c) Profit $P\Delta x - \Delta y^* = 6713$ USD; without the fee 7147 USD.
- Interpretation: arbitrage stops when the marginal cost of Ether in the pool reaches P ; the pool price ends inside its new band, $P \in [\gamma p', p'/\gamma] = [3280.67, 3300.45]$, not at P ; the profit is paid by the liquidity providers, who sold Ether below the market price.

A4: Arbitrage in a Weighted Pool [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Weighted pool (Balancer type) $x^w y^{1-w} = k$ with $w = 0.8$, $x = 1\,000$ Ether and pool price $p = 3\,000$; fee 0.3% on the input
- ▶ The outside price rises to $P = 3\,300$; one arbitrage trade, no gas cost

□ Tasks

- ▶ (a) Differentiate the invariant to show that the pool price is $p = \frac{w}{1-w} \frac{y}{x}$, and compute y .
- ▶ (b) Show that the no-arbitrage band $\gamma P \leq p \leq P/\gamma$ does not depend on w .
- ▶ (c) Derive the reserves after the optimal arbitrage (the fee stays in the pool, as in A3), compute Δy^* , the profit and the profit per unit of pool value (the pre-trade value $px + y$ at the pool price; A3: 6 million USD), and compare with A3.

□ Report

- ▶ y , x' , the effective and actual y' , Δy^* , the pool value, the profit, the profit per unit of value and the ratio to A3
- ▶ Model: A3

□ Interpretation question

- ▶ Why does a pool with $w = 0.8$ lose less to arbitrageurs than a 50/50 pool?

A5: Impermanent Loss of a Weighted Pool [Solved]

Task

▣ Data and assumptions

- ▶ A liquidity provider owns a pool $x^w y^{1-w} = k$; arbitrage keeps the pool at the market price P (A4, without fees)
- ▶ The price moves from P_0 to $P_1 = rP_0$; the benchmark is holding the initial tokens

▣ Tasks

- ▶ (a) Show that the value shares stay w and $1 - w$ and that $V(P) = cP^w$.
- ▶ (b) Derive $IL_w(r) = r^w / (wr + 1 - w) - 1$ and check that $w = 1/2$ gives $2\sqrt{r}/(1 + r) - 1$.
- ▶ (c) Expand $\ln(1 + IL_w)$ to second order in $u = \ln r$ and compute IL_w for $r = 2$ and $r = 4$, $w = 0.5$ and 0.8 .

▣ Report

- ▶ The formula, the approximation and a table of four values

▣ Interpretation question

- ▶ Which pool loses less when the price doubles, and why?

A5: Impermanent Loss of a Weighted Pool (Solution 1/2) [Solved]

- (a) From A4(a) the pool price equals P : $Px = \frac{w}{1-w}y$, so $Px = wV$ and $y = (1-w)V$ with $V = Px + y$.
- Substituting into the invariant: $(wV/P)^w((1-w)V)^{1-w} = k$, so $V = k w^{-w}(1-w)^{w-1}P^w = cP^w$.
- (b) Pool: $V_1/V_0 = r^w$; holding: $V_H/V_0 = wr + (1-w)$; so $IL_w(r) = r^w/(wr + 1 - w) - 1$; at $w = 1/2$: $\sqrt{r}/[(r+1)/2] - 1 = 2\sqrt{r}/(1+r) - 1$.

A5: Impermanent Loss of a Weighted Pool (Solution 2/2) [Solved]

- (c) $\ln(1 + \text{IL}_w) = wu - \ln(1 + w(e^u - 1))$; with $e^u - 1 = u + u^2/2$ and $\ln(1 + z) = z - z^2/2$: $\ln(1 + \text{IL}_w) \approx -\frac{1}{2}w(1 - w)u^2$.
- $r = 2$: -5.72% for $w = 0.5$ (approximation -6.01%) and -3.27% for $w = 0.8$ (-3.84%).
- $r = 4$: -20.00% ($w = 0.5$) and -10.84% ($w = 0.8$); the approximation is good for moderate moves only.
- Interpretation: the loss scales with $w(1 - w)$, the curvature of $V(P)$; a pool that holds mostly Ether rebalances less against the trend and sells less volatility.

A6: Loss Versus Rebalancing by Itô [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Price $dP_t = \mu P_t dt + \sigma P_t dW_t$; pool value $V(P) = cP^w$ (A5); frictionless arbitrage, no fees [Milionis et al. \(2022\)](#)
- ▶ The rebalancing portfolio holds at every instant the same quantity of Ether as the pool, $V'(P_t)$, and trades at market prices

□ Tasks

- ▶ (a) Compute V' and V'' , apply Itô's lemma to $V(P_t)$, and show that $d(R_t - V_t) = \frac{1}{2}w(1-w)\sigma^2 V_t dt$.
- ▶ (b) Check that $w = 1/2$ gives the lecture's rate $\sigma^2/8$.
- ▶ (c) Compute the annual LVR rate for $\sigma = 70\%$ and $w = 0.5, 0.8$ and 0.95 .

□ Report

- ▶ The derivation and three rates in % a year
- ▶ Model: A5

□ Interpretation question

- ▶ Why does the drift μ not appear in the loss?

A7: Threshold AR Algebra [Solved]

Task

▣ Data and assumptions

- ▶ Peg deviation in basis points: $d_t = \phi_{\text{in}} d_{t-1} + \varepsilon_t$ if $|d_{t-1}| \leq c$, and $d_t = \phi_{\text{out}} d_{t-1} + \varepsilon_t$ otherwise Balke & Fomby (1997)
- ▶ USD Coin estimates from B3: $c = 2.8650$, $\phi_{\text{in}} = 0.5855$, $\phi_{\text{out}} = 0.2832$; use the skeleton $\varepsilon_t = 0$

▣ Tasks

- ▶ (a) Derive the half-life in each regime and compute it.
- ▶ (b) Starting from $d_0 = -285$ bp (the close of 11 March 2023), derive and compute the number of days until the deviation is back inside the band.
- ▶ (c) State what $\phi_{\text{in}} = 1$ would mean economically.

▣ Report

- ▶ Two half-lives in days and the number of days of (b)

▣ Interpretation question

- ▶ What does $\phi_{\text{in}} < 1$ say about who trades inside the band?

A7: Threshold AR Algebra (Solution) [Solved]

- (a) While the process stays in one regime, $d_h = \phi^h d_0$; $\phi^h = 0.5$ gives $h_{1/2} = \ln 0.5 / \ln \phi$.
- Inside: $\ln 0.5 / \ln 0.5855 = 1.29$ days; outside: $\ln 0.5 / \ln 0.2832 = 0.55$ days.
- (b) Outside the band $|d_n| = |d_0| \phi_{\text{out}}^n \leq c \iff n \geq \ln(c/|d_0|) / \ln \phi_{\text{out}} = \ln(2.8650/285) / \ln 0.2832 = 3.65$.
- So after 4 days the deviation is inside the band; from then on the slower inside dynamics apply.
- (c) $\phi_{\text{in}} = 1$ is a random walk inside the band, the band-TAR case of [Balke & Fomby \(1997\)](#): no one trades until arbitrage pays.
- Interpretation: the estimate $\phi_{\text{in}} < 1$ means market makers pull the price towards par even inside the band.

A8: What Does a Linear AR(1) Estimate? [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ The USD Coin deviations of A7 and B3; a linear AR(1) $d_t = \phi d_{t-1} + e_t$ fitted by OLS without an intercept
- ▶ $\hat{\phi}_{\text{in}}$ and $\hat{\phi}_{\text{out}}$ are the regime-wise OLS estimates at the threshold $\hat{c}$ of B3

□ Tasks

- ▶ (a) Show that, in sample, $\hat{\phi}_{\text{lin}} = (1 - s)\hat{\phi}_{\text{in}} + s\hat{\phi}_{\text{out}}$ with $s = \sum_{\text{out}} d_{t-1}^2 / \sum_t d_{t-1}^2$.
- ▶ (b) Compute s for USD Coin and check the identity numerically against $\hat{\phi}_{\text{lin}}$ from B3.
- ▶ (c) Compute the linear half-life and say which regime it describes.

□ Report

- ▶ s , the weighted average, $\hat{\phi}_{\text{lin}}$ and the linear half-life
- ▶ Model: A7

□ Interpretation question

- ▶ Why can a linear AR(1) report “fast reversion” although most days show slow reversion?

B1: Bitcoin's Volatility and the Weekend Effect [Solved]

Task

▣ Data and assumptions

- ▶ Daily log returns of Bitcoin (BTC-USD.CC), 2 January 2018 – 18 September 2026, 3 182 days, 7 days a week; S&P 500 (GSPC.INDX) on its own calendar
- ▶ Split at the first spot ETF day, 11 January 2024: 2 200 days before, 982 after

▣ Tasks

- ▶ (a) Compute the annualised volatility $s\sqrt{365}$ with a 95% moving-block bootstrap interval (blocks of 20 days, 2 000 resamples) [Künsch \(1989\)](#).
- ▶ (b) Compute the ratio of the weekend to the weekday variance of returns before and after the ETFs, each with a block-bootstrap interval (blocks of 21 days).
- ▶ (c) Compute a 95% interval for the change in the ratio from the difference of the two bootstrap distributions.

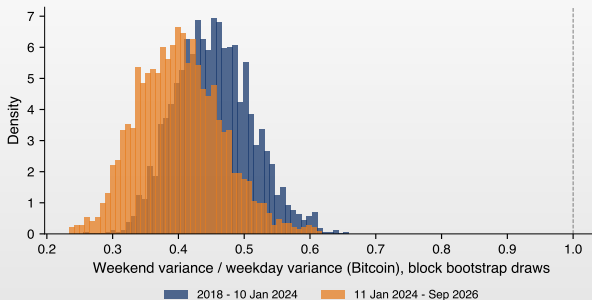
▣ Report

- ▶ Volatility with CI and the ratio to the S&P 500; both ratios with CIs; the change with CI; the bootstrap histogram

▣ Interpretation question

- ▶ Is the weekend lull a new, ETF-driven effect? Why blocks and not single days?

B1: Bitcoin's Volatility and the Weekend Effect (Solution 1/2) [Solved]



- Block-bootstrap distributions of the weekend / weekday variance ratio; the dashed line marks a ratio of 1

B1: Bitcoin's Volatility and the Weekend Effect (Solution 2/2) [Solved]

- (a) Steps: compute s , resample 2 000 series of 20-day blocks, recompute $s\sqrt{365}$, take the 2.5% and 97.5% percentiles.
- Result: 64.1%, 95% CI [58.2, 70.4]; S&P 500 19.2%, a ratio of 3.3; excess kurtosis 14.9.
- (b) Before: 0.45 [0.35, 0.57]; after: 0.39 [0.29, 0.53].
- (c) Change -0.05 , 95% CI $[-0.22, 0.11]$: the interval contains 0, so at the 5% level we do not reject an unchanged weekend effect.
- Interpretation: weekends have less than half the weekday variance before and after 2024, so the lull is not ETF-driven; blocks keep the volatility clusters that single-day resampling would destroy.
- Caveat: these percentile intervals for a variance (and a variance ratio) assume $Er^4 < \infty$; with $\hat{\alpha}_H$ near 2.7 (lecture) this is doubtful and the bootstrap need not be consistent [Athreya \(1987\)](#): treat the intervals as indicative (an m -out-of- n bootstrap is the robust alternative).

B2: Ether and Solana [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily log returns of Ether (ETH-USD.CC, 2018–2026) and Solana (SOL-USD.CC, April 2020 – 2026); S&P 500 over the same periods
- ▶ Same split date (11 January 2024) and bootstrap settings as B1

▣ Tasks

- ▶ (a) Compute each coin's annualised volatility with a block-bootstrap CI and its ratio to the S&P 500 volatility.
- ▶ (b) Compute the weekend / weekday ratio before and after the ETFs and a CI for the change.

▣ Report

- ▶ One table with both coins and Bitcoin from B1
- ▶ Model: B1

▣ Interpretation question

- ▶ Which coin is the riskiest relative to shares? Is the weekend effect weaker for coins with more on-chain activity?

B3: Is the USD Coin Peg a Threshold Process? (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Daily close of USD Coin (USDC-USD.CC), 1 January 2021 – 18 September 2026, 2 087 days; deviation $d_t = 10^4(P_t - 1)$ in basis points
- ▶ Model of A7 (no intercept); errors may be heteroskedastic; test level 5%

▣ Tasks

- ▶ (a) Compute the standard deviation, the number of days with $|d_t| > 50$ bp, the lowest close and the lowest daily low.
- ▶ (b) Estimate c by least squares over the grid of $|d_{t-1}|$ values that leaves at least 15% of the days in each regime, and estimate $\phi_{\text{in}}, \phi_{\text{out}}$ with White standard errors.
- ▶ (c) Test $H_0: \phi_{\text{in}} = \phi_{\text{out}}$ with $\sup_c W(c)$ and the fixed-regressor bootstrap Hansen (1996): $d_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0, 1)$, $\hat{e}_t$ the residuals of the linear AR(1) under H_0 , regressors d_{t-1} fixed, 999 draws; also report the χ^2_1 p -value.
- ▶ (d) Repeat (b)–(c) without 9–16 March 2023: drop every pair (d_{t-1}, d_t) that touches these dates, so that no pair spans the gap (8 and 17 March are not consecutive).

B3: Is the USD Coin Peg a Threshold Process? (2/2) [Solved]

Task

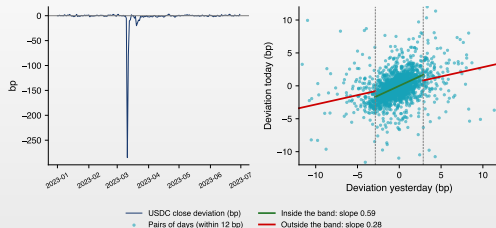
☐ Report

- ▶ $\hat{c}$, both slopes with SE, half-lives, sup W , both p -values, the scatter chart

☐ Interpretation question

- ▶ Why is χ^2_1 wrong here, and does one stress week carry the evidence of a band?

B3: Is the USD Coin Peg a Threshold Process? (Solution 1/3) [Solved]



- Left: close deviation in the first half of 2023; right: today against yesterday within ± 12 bp, with the fitted slopes inside and outside $\pm \hat{c}$ (dashed lines)
- (a) s.d. 7.5 bp (without March 2023: 3.7); 3 days beyond 50 bp; lowest close -285 bp and lowest low -1226 bp, both on 11 March 2023.

B3: Is the USD Coin Peg a Threshold Process? (Solution 2/3) [Solved]

- (b) Steps: sort the days by $|d_{t-1}|$; for each c on the grid $[0.3, 2.9]$ fit OLS in each regime and keep the c with the smallest total sum of squared residuals.
- Result: $\hat{c} = 2.9$ bp, 15.7% of days outside; $\hat{\phi}_{\text{in}} = 0.59$ (SE 0.04), $\hat{\phi}_{\text{out}} = 0.28$ (SE 0.03); half-lives 1.29 and 0.55 days.
- (c) $W(c) = (\hat{\phi}_{\text{in}} - \hat{\phi}_{\text{out}})^2 / (\hat{v}_{\text{in}} + \hat{v}_{\text{out}})$ on each c ; $\sup W = 32.5$.
- Bootstrap: regress d_t^* on the same fixed d_{t-1} , recompute $\sup W^*$ 999 times; $p = 0.001$ (χ_1^2 : < 0.001); decision: reject linearity at 5%.

B3: Is the USD Coin Peg a Threshold Process? (Solution 3/3) [Solved]

- (d) Without the pairs touching 9–16 March 2023: $\sup W = 5.5$, bootstrap $p = 0.291$: linearity is no longer rejected; χ_1^2 would give 0.019, a rejection by a miscalibrated test.
- Why χ_1^2 fails: c is not identified under H_0 , and the supremum over many c is larger than any single $W(c)$ (the Davies problem).
- Why a wild bootstrap: the excess kurtosis is 1031; multiplying each residual by η_t keeps its own variance, which an i.i.d. resampling of residuals would mix across calm and stress days.
- Economic conclusion: the evidence for nonlinear adjustment is sensitive to the stress week; the remaining sample does not reject a linear, fast-reverting AR (non-rejection, not proof of linearity).

B4: Tether and Dai [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily closes of Tether (USDT-USD.CC) and Dai (DAI-USD.CC), 1 January 2021 – 18 September 2026; deviations in bp
- ▶ Hypothesis to discuss: Tether has the widest band, because fewer traders can redeem directly with the issuer; the tests below are tests of linearity, not of differences between bands

▣ Tasks

- ▶ (a) Estimate the threshold AR of B3 for each coin: $\hat{c}$, $\hat{\phi}_{\text{in}}$, $\hat{\phi}_{\text{out}}$ with White SE.
- ▶ (b) Test linearity with sup W and the fixed-regressor bootstrap (999 draws), with and without 9–16 March 2023, at the 5% level.

▣ Report

- ▶ One table with Tether, Dai and USD Coin (B3)
- ▶ Model: B3

▣ Interpretation question

- ▶ Which estimated band is widest, and for which coins does the bootstrap reject linearity? What would a test of the difference between two thresholds need? Which of the three coins reverts fastest outside its band?

B5: IBIT: Tracking, Timing and the Fee Drift [Solved]

Task

▣ Data and assumptions

- ▶ IBIT (IBIT.US, adjusted close) and Bitcoin (BTC-USD.CO) on IBIT's trading days, 11 January 2024 – 18 September 2026: 673 days and 140 Friday-to-Friday weeks; prices joined first, then log returns
- ▶ Sponsor fee 0.25% a year paid in Bitcoin [BlackRock \(2026\)](#); level 5%

▣ Tasks

- ▶ (a) Estimate the daily and the weekly beta with Newey–West SE [Newey & West \(1987\)](#), and the Dimson sum beta $b_{-1} + b_0 + b_1$ with one lag and one lead [Dimson \(1979\)](#).
- ▶ (b) Run an ADF test (with trend, lags by AIC) [Dickey & Fuller \(1979\)](#) and a KPSS test (with trend) [Kwiatkowski et al. \(1992\)](#) on $\ell_t = \ln(P_t^{\text{IBIT}} / P_t^{\text{BTC}})$.
- ▶ (c) Estimate the annual fee drift as the mean of $\Delta \ell_t$ (HAC SE, 6 lags) and as the slope of ℓ_t on time in years (HAC SE, 20 lags), each with a 95% CI.

▣ Report

- ▶ Three betas with SE, the two test statistics with decisions, both drifts with CIs

▣ Interpretation question

- ▶ Why is the daily OLS beta below 1? Which drift estimator is valid and which is precise?

B5: IBIT: Tracking, Timing and the Fee Drift (Solution 1/2) [Solved]

- (a) Daily OLS $\hat{\beta} = 0.93$; weekly $\hat{\beta} = 1.007$ (SE 0.011), $R^2 = 0.98$.
- Dimson: $\hat{b}_{-1} = 0.05$, $\hat{b}_0 = 0.94$, $\hat{b}_1 = 0.01$; sum 1.00 (SE 0.023), $t(\beta = 1) = -0.07$: not different from 1.
- Reading: part of each Bitcoin move reaches IBIT a day later; the lag term restores $\beta = 1$ on daily data, as the weekly regression does.
- (b) ADF = -11.07 , $p < 0.001$: reject the unit root at 5%; KPSS = 0.041, $p > 0.10$: do not reject trend stationarity.

B5: IBIT: Tracking, Timing and the Fee Drift (Solution 2/2) [Solved]

- (c) Mean of $\Delta\ell_t$ times days per year: -0.36% , 95% CI $[-9.28, 8.57]$.
- Trend slope: -0.237% a year, 95% CI $[-0.315, -0.158]$: excludes 0 and contains -0.25% .
- Why they differ: the mean of $\Delta\ell_t$ telescopes to $(\ell_T - \ell_0)/T$, so it uses only two noisy days; the trend slope uses every day.
- Under the maintained model $\ell_t = \text{fee trend} + \text{stationary timing noise}$ (ADF rejects, KPSS does not), the trend slope with HAC SE is consistent and its CI is valid; the mean of $\Delta\ell_t$ is also consistent, but $\Delta\ell_t$ then has zero long-run variance, so its HAC interval is poorly calibrated.
- With a unit root plus drift, the trend slope still estimates the drift, but the usual trend HAC interval is invalid; the mean of differences then has the standard CI. Choosing by a pretest does not guarantee post-selection coverage.
- Economic conclusion: the fee is visible in the data at the 0.25% rate, and IBIT tracks Bitcoin one for one.

B6: ETHA and Ether [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ ETHA (ETHA.US, adjusted close) and Ether (ETH-USD.CC) on ETHA's trading days, 23 July 2024 – 18 September 2026: 541 days, 112 weeks
- ▶ Sponsor fee 0.25% a year; level 5%

▣ Tasks

- ▶ (a) Estimate the daily, weekly and Dimson betas as in B5(a).
- ▶ (b) Run the ADF and KPSS tests of B5(b) on the log price ratio.
- ▶ (c) Estimate both drifts of B5(c) with 95% CIs.

▣ Report

- ▶ Betas with SE, tests with decisions, both drifts with CIs, tracking errors
- ▶ Model: B5

▣ Interpretation question

- ▶ Is ETHA a worse tracker than IBIT, or only noisier? How many years would pin the fee down to $\pm 0.05\%$?

B7: When Did Strategy's Bitcoin Beta Change? (1/2) [Solved]

Task

▣ Data and assumptions

- ▶ Weekly (Friday) log returns of Strategy (MSTR.US, formerly MicroStrategy), Bitcoin and the S&P 500, 16 April 2021 – 18 September 2026, 284 weeks; prices joined first
- ▶ Model $r_t^{\text{MSTR}} = a + \beta_{\text{BTC}} r_t^{\text{BTC}} + \beta_{\text{SPX}} r_t^{\text{SPX}} + u_t$; level 5%

▣ Tasks

- ▶ (a) Estimate the model with Newey–West SE on the full period and on the two subperiods split at the first spot ETF day (weeks up to 5 January 2024; from 12 January 2024), and test $\beta_{\text{BTC}} = 1$ in each.
- ▶ (b) Test for a break in all three coefficients at an unknown date with sup-Wald (HAC covariance in each regime, 15% trimming) [Andrews \(1993\)](#); take the 5% critical value from the simulated limit $\sup_r \|B(r) - rB(1)\|^2 / [r(1-r)]$, B three-dimensional.
- ▶ (c) Estimate the break date $\hat{T}$ by least squares and its 95% CI [Bai \(1997\)](#): $\hat{T} \pm (\lceil 11/\hat{L} \rceil + 1)$ weeks, $\hat{L} = (\hat{\delta}' \hat{Q} \hat{\delta})^2 / (\hat{\delta}' \hat{\Omega} \hat{\delta})$, where $\hat{\delta}$ is the post-minus-pre coefficient vector, $\hat{Q} = X'X/T$, and $\hat{\Omega}$ the Bartlett long-run covariance of $x_t \hat{u}_t$ (Newey–West lags, residuals of the two-regime fit).
- ▶ Assumptions: a small shift, Q and Ω the same in both regimes; 11 is the 97.5% quantile of $\arg \max_s \{W(s) - |s|/2\}$, W a two-sided Brownian motion, so the interval is symmetric.

B7: When Did Strategy's Bitcoin Beta Change? (2/2) [Solved]

Task

☐ Report

- ▶ The table of (a), sup W with the critical value and the decision, the date with its CI and the coefficients in each regime

☐ Interpretation question

- ▶ Do the data select January 2024? Why is a split at an imposed date not a test of when the beta changed?

B7: When Did Strategy's Bitcoin Beta Change? (Solution 1/2) [Solved]

	n	$\hat{\beta}_{BTC}$ (SE)	$t(\beta_{BTC} = 1)$	$\hat{\beta}_{SPX}$ (SE)	alpha, % a year (t)
□ Apr 2021 – Sep 2026	284	1.18 (0.11)	1.64	1.02 (0.24)	−4.5 (−0.19)
□ Apr 2021 – 5 Jan 2024	143	1.01 (0.14)	0.08	1.40 (0.35)	−0.7 (−0.02)
□ 12 Jan 2024 – Sep 2026	141	1.50 (0.12)	4.12	0.48 (0.29)	−9.3 (−0.29)

- (a) The imposed split suggests a change: $\hat{\beta}_{BTC}$ from 1.01 to 1.50 ($t = 4.12$ against 1 after 2024); alpha is never significant.

B7: When Did Strategy's Bitcoin Beta Change? (Solution 2/2) [Solved]

- (b) Steps: for every candidate week in the middle 70% fit both regimes, compute $W(\tau)$ with HAC covariances and take the maximum.
- Result: $\sup W = 35.49$ against a 5% critical value of 13.95, $p < 0.001$: reject stability at 5%.
- (c) Break date 24 Jun 2022, 95% CI 31 Dec 2021 – 16 Dec 2022; W at the ETF launch only 8.64.
- Regimes: Bitcoin beta $0.69 \rightarrow 1.48$; S&P 500 beta $1.25 \rightarrow 1.00$.
- Interpretation: the data choose mid-2022, a period of market-wide crypto stress, not the ETF launch; the date alone does not identify the cause; an imposed split finds a change wherever the true break lies before it, so it cannot date the change.

B8: Coinbase [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Weekly log returns of Coinbase (COIN.US, listed April 2021), Bitcoin and the S&P 500, 16 April 2021 – 18 September 2026, 283 weeks
- ▶ Same model, settings and level as B7

□ Tasks

- ▶ (a) Estimate the two-factor model on the full period and the two subperiods with Newey–West SE.
- ▶ (b) Run the sup-Wald test for a break at an unknown date.
- ▶ (c) Estimate the break date and its 95% CI, and the coefficients in each regime.

□ Report

- ▶ Table of (a), the test with its decision, the date with CI
- ▶ Model: B7

□ Interpretation question

- ▶ Is Coinbase a Bitcoin proxy or a US growth stock? Do Strategy and Coinbase break in the same week, and what common event could explain it?

C1: Alternative Asset, Risk Asset or Hedge? (1/2) [Proposed]

Solution discussed in class

Task

▣ Data and assumptions

- ▶ Daily log returns of Bitcoin, the S&P 500, Nasdaq 100 (QQQ), gold (GLD) and long Treasuries (TLT) on common days (prices joined first), January 2021 – 10 January 2024 against 11 January 2024 – 18 September 2026
- ▶ Definitions of hedge and safe haven: [Baur & Lucey \(2010\)](#); earlier evidence for Bitcoin: [Bouri et al. \(2017\)](#); level 5%

▣ Tasks

- ▶ (a) Compute the correlations with Bitcoin in both periods, with moving-block bootstrap CIs (20-day blocks, 2 000 draws) for each and for the change.
- ▶ (b) Test $H_0: \Delta\rho = 0$ for the S&P 500 with the Forbes–Rigobon adjusted correlation [Forbes & Rigobon \(2002\)](#) and a block-bootstrap CI.
- ▶ (c) Estimate the hedge / safe-haven regression with interactions at the 10%, 5% and 1% S&P 500 quantiles and Newey–West SE, and the quantile-regression betas $\beta(\tau)$ at $\tau = 1\%, 5\%, 50\%$ [Koenker & Bassett \(1978\)](#) with block-bootstrap SE.
- ▶ (d) Compare downside and upside betas and Bitcoin's return on the ten worst S&P 500 days.

C1: Alternative Asset, Risk Asset or Hedge? (2/2) [Proposed]

Solution discussed in class

Task

☐ Report

- ▶ A correlation table, the test of (b), the regression and quantile betas, a two-slide verdict linked to the distribution features of the lecture ([Pele et al. \(2023\)](#))

☐ Interpretation question

- ▶ Is Bitcoin “alternative” in shape but “risk” in co-movement? Did the ETFs change that?

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Compare Bitcoin with lending USD Coin (USDC) at 5% a year: compute Bitcoin’s annualised Sharpe ratio on daily data for 2021–2025, with the USDC yield as the risk-free rate”
- The assistant answers with this code:

```
import numpy as np, pandas as pd
btc = pd.read_csv('data/market/BTC-USD.CC.csv', index_col='date',
                  parse_dates=True)['close']
r = np.log(btc.loc['2021':'2025']).diff().dropna()    # daily log returns
mu = r.mean() * 252                                  # annual mean
sd = r.std() * np.sqrt(252)                           # annual volatility
rf = 0.05                                             # USDC lending: risk-free
sharpe = (mu - rf) / sd
print(f"mu={mu:.1%}, vol={sd:.1%}, Sharpe={sharpe:.2f}")
```

- “Output: mu = 15.1%, vol = 48.5%, Sharpe = 0.21. USDC is fully backed by dollars and has never traded below \$0.99, so lending it is a risk-free investment”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- The answer contains several substantive errors, in the code and in the text: find all of them
- For each error
 - ▶ say what is wrong and why
 - ▶ say how you would detect it: a count in the data, the data themselves, a definition from the lecture
 - ▶ give the corrected line or statement
- Rerun the corrected code (returns computed before selecting 2021–2025, simple excess returns, r_f from Treasury bills); report the mean, the volatility and the Sharpe ratio
- Log the errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- How many daily returns should 2021–2025 contain for Bitcoin? How many does the code produce, and why?
- Which of the errors can you check in one line with the course data?
- If lending USDC were risk-free, why would it pay more than a Treasury bill?

C3: Is Crypto Momentum Still There? (1/2) [Proposed]

Solution discussed in class

Task

□ Data and assumptions

- ▶ Context: [Liu & Tsyvinski \(2021\)](#) (lecture case study) find little exposure of crypto returns to stock factors but weekly time-series momentum, in a sample ending on 31 May 2018 [Liu & Tsyvinski \(2018\)](#)
- ▶ Daily closes of Bitcoin, XRP and Ether from data/market (weekly: Sunday to Sunday; monthly); five factors [Fama & French \(2015\)](#), momentum and the risk-free rate from the [Kenneth French Data Library](#); in sample as published (Bitcoin from 1 January 2011, XRP from 4 August 2013, Ether from 7 August 2015, to 31 May 2018), out of sample 1 June 2018 – 13 September 2026; level 5%

□ Tasks

- ▶ (a) For Bitcoin in the published period, rebuild Table 1: weekly and monthly moments, Sharpe ratio, and the daily counts beyond $\pm 5\%$, $\pm 10\%$, $\pm 20\%$; compare them with the paper.
- ▶ (b) Estimate the monthly CAPM and five-factor regressions of Table 5 for Bitcoin in both periods; report α , the market β and R^2 , each t with usual and bootstrapped standard errors.
- ▶ (c) For each coin and period, estimate $r_{t+h} = a + b r_t + e_{t+h}$ on weekly returns, $h = 1, \dots, 4$ (Tables 14, 18), and the quintile sort with cut-offs from the first two years (Table 17); report the top-minus-bottom difference.
- ▶ (d) Test whether the one-week slope b of Bitcoin changed: an out-of-sample interaction with HAC errors [Newey & West \(1987\)](#) and a sup-Wald test at an unknown date [Andrews \(1993\)](#), as in B7.

C3: Is Crypto Momentum Still There? (2/2) [Proposed]

Solution discussed in class

Task

▣ Report

- ▶ A table paper – replication – out of sample for (a) and (b), a table of slopes with both t -statistics for (c), the break test with its estimated date, and the full list of tests run
- ▶ Model: B1, B7

▣ Interpretation question

- ▶ If weekly momentum disappears after 2018 while the stock beta rises, has crypto become a conventional asset, or was the 2011–2018 momentum a small-sample result? Which of your numbers separates the two readings?

Common Pitfalls

- ▣ Forgetting the divisor when an index changes constituents or supplies
- ▣ Using the pool price as the execution price: large trades pay the price impact
- ▣ Judging a liquidity position against holding instead of against rebalancing
- ▣ Testing a threshold or a break date with χ^2 critical values; imposing the break date instead of estimating it
- ▣ Daily OLS betas between assets that close at different times (use Dimson); a trend regression without a unit-root test
- ▣ Calling Bitcoin a hedge without testing the extreme-day betas

Take-Home Assignment (Team Project)

- ▣ Choose three crypto-assets not analysed in class (or a stablecoin and two coins)
- ▣ Build a market-value weighted index with a divisor and compare it with Bitcoin: return, volatility, tracking error, with bootstrap intervals
- ▣ Run the hedge / safe-haven regression of Part C for your index against the S&P 500 and gold
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations
- ▣ Log every AI error you catch in `AI_ERRORS.md`, as in C2: what was asked, what was wrong, how you found out, the correct version

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Are Cryptos becoming alternative Assets?
- ▣  Comparison of Cryptocurrency Indices
- ▣  Stabilising Onchain Stablecoin Risks
- ▣  On Crypto-backed Loans

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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